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Comparing human and synthetic data in service research: using augmented language models to study service failures and recoveries

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Abstract

Purpose – This study aims to explore Augmented Language Models (ALMs) for synthetic data generation in services marketing and research. It evaluates ALMs' potential in mirroring human responses and behaviors in service scenarios through comparative analysis with five empirical studies.

Design/methodology/approach – The study uses ALM-based agents to conduct a comparative analysis, leveraging SurveyLM (Bickley *et al.*, 2023) to generate synthetic responses to the scenario-based experiment in Söderlund and Oikarinen (2018) and four more recent studies from the *Journal of Services Marketing*. The main focus was to assess the alignment of ALM responses with original study manipulations and hypotheses.

Findings – Overall, our comparative analysis reveals both strengths and limitations of using synthetic agents to mimic human-based participants in services research. Specifically, the model struggled with scenarios requiring high levels of visual context, such as those involving images or physical settings, as in the Dootson *et al.* (2023) and Srivastava *et al.* (2022) studies. Conversely, studies like Tariq *et al.* (2023) showed better alignment, highlighting the model's effectiveness in more textually driven scenarios.

Originality/value – To the best of the authors' knowledge, this research is among the first to systematically use ALMs in services marketing, providing new methods and insights for using synthetic data in service research. It underscores the challenges and potential of interpreting ALM versus human responses, marking a significant step in exploring AI capabilities in empirical research.

Keywords Large language models, Augmented language models, Service research, Scenario-based methods, Synthetic data generation, Service failures and recoveries

Paper type Research paper

1. Introduction

Service researchers use multiple different paradigms to capture the dynamic nature of the field, ranging from quantifiable measures of variables (i.e. positivist approach) through to the social construction of meaning in service interactions (i.e.

Availability of data and materials: Data and materials used in the study are available at: <https://osf.io/b4udp/>.

Competing interests: SJB, HFC, BD, BT and ST are co-founders of the company/research lab that developed SurveyLM, and therefore, may appear or be perceived to have competing interests.

The authors extend their gratitude to the guest editors for their dedicated time, support and effort in developing this special issue, and authors thank the two anonymous reviewers for their constructive feedback and valuable criticism.

Use of Artificial Intelligence (AI) and AI-assisted Technologies: This paper incorporated the use of AI-assisted technologies (e.g., GPT-3.5-turbo, GPT-3.5-turbo-16k, GPT-4 and GPT-4o), specifically for recommending text edits to enhance readability and language quality. These tools were used during the manuscript preparation (Oct to Nov 2023 and Jul to Aug 2024). All content has been critically reviewed and curated by the authors. This disclosure affirms the responsible use of AI.

The current issue and full text archive of this journal is available on Emerald Insight at: <https://www.emerald.com/insight/0887-6045.htm>



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hermeneutic approach) (Tronvoll *et al.*, 2011). Although researchers typically adopt methodologies that reflect their epistemological approach or phenomena of interest, in response to calls for more refined and evolved methodologies to capture the complexity and detail required in services research (Tronvoll *et al.*, 2011; Kuppelwieser and Klaus, 2023), this paper investigates the value of *augmented language models* (ALMs, see, e.g. Mialon *et al.*, 2023) [1] for service research. Specifically, we explore the utility of ALMs by comparing human and synthetic agent responses to service failure and recovery scenarios from five empirical studies in the services marketing literature. We leverage *SurveyLM*, an ALM-based simulation platform developed by some of the authors of this paper (Bickley *et al.*, 2023), to conduct these comparisons.

Service failures can be described as errors within a service experience, many of which occur due to the heterogeneity and variability characterizing services (Hess, 2008). In response, service recovery encompasses the efforts and actions taken by a service provider (both at the firm or employee level) to respond to consumer complaints, often with the goal of minimizing switching intentions or ensuring consumer satisfaction (Spreng *et al.*, 1995). Service failure and recovery processes form part of a substantial body of research within the entirety of the service discipline, with Hart *et al.* (1990) even suggesting in as early as 1990 that the processes related to service recovery are a “profitable art”. In *Journal of Services Marketing* (JSM) alone, the topic of service failure, complaining and recovery is among the most prominent clusters of research, with continuing interest in service recovery and service encounters (Donthu *et al.*, 2022).

Recent advances in ALMs provide a unique opportunity to study service failures by testing a rich set of “break the script” variables that offer insights to improve service experiences. As with any complex encounter involving human behaviors, there are an extensive number of variables (both service specific and intrinsic to an individual) that may impact end outcomes (e.g. satisfaction, service evaluations, post-purchase behaviors). Often due to limited resources and time constraints, both researchers and service practitioners are required to narrow the scope of their exploration into the impacts of such variables.

We thereby propose that ALMs may provide an accelerated modality by which researchers and practitioners can gain a preliminary understanding of the impacts of a broader range of variables upon their intended end outcome, due to the competitive cost, speed of changes and the richness to explore the relevance of contextual factors in a comparative way. For example, one might be interested in exploring the impacts of joke-telling in service encounters upon customer satisfaction in service contexts different to those explored in the current study (e.g. repeated customers and long-term service relationships). In other words, ALMs show potential in providing the modality to expand from assessing this impact upon a single variable like gender, culture and age and rather provide an iterative rapid way of manipulating specific contextual variables whilst keeping all other elements constant. Thereby, allowing researchers to rapidly test hypotheses and explore a wider range of variables than would be possible with human participants alone. This approach not only accelerates the research process but also identifies new areas for future in-depth human studies, highlighting avenues to broaden the scope of original studies.

ALMs are also particularly suited to research in service failure and service recovery due to the prevalence of vignettes or scenario-based decision-making tasks in this literature area. Service employees often need to use their professional expertise, creative thinking and discretion to accommodate or overcome unusual problems (Hart *et al.*, 1990). Scholars have tested a wide range of variables, including how variations in an employee’s cultural values, empowerment, emotional intelligence and managerial involvement impact recovery measures (Lin, 2010). A service recovery system involves a variety of different relationships that are open to empirical examination, with research exploring everything from the antecedents and consequences of an individual employee’s recovery performance, through to the drivers and outcomes of customer satisfaction under different scenarios (high/low effort, empathetic/neutral, fast/slow) of service and recovery efforts (Van Vaerenbergh and Orsingher, 2016). This shows the numerous avenues for empirical research using ALMs in service failure and recovery, particularly for those operating within the dominant positivist paradigm of service research.

This research aims to understand the usefulness of ALMs as an additional methodological toolbox for service researchers. Before we can integrate ALMs as potential generative research tools or substitutes for human research participants in the service discipline, we must first establish the validity and reliability of ALMs to act in human-like ways. As an initial step toward this objective, we compare responses from humans and synthetic agents to service failure and recovery scenarios extracted from five empirical studies which offer a diverse range of service contexts and other elements of heterogeneity, such as primary mode(s) of communication (e.g. textual, visual), to understand the capabilities and limitations of ALMs. This includes from retail and financial services to intercultural encounters. Therefore, providing a comprehensive overview of different service failure and recovery scenarios.

The five studies serve as benchmarks for our analysis which focuses on evaluating the alignment between human and ALM-generated responses and identifying the mechanisms driving any discrepancies. The synthetic agent simulations use primary materials as closely as possible to the original studies, embedding demographic attributes to each individual agent that reflect the distribution(s) of demographic characteristics of the original sample populations. Therefore, we do expect some differences (and also some similarities) between the human and synthetic responses of the selected studies, which we discuss later in the paper.

Although the utility of ALMs and *SurveyLM* can be assessed in diverse service settings and at different levels of a multi-level service system, this paper restricts its initial exploration to four recent papers (published in 2022–2023) from the *Journal of Services Marketing* and one replication study (published in, 2018) from the *Journal of Retailing and Consumer Services*. These papers, focusing on service failure or recovery using scenario-based tasks, provide a targeted yet detailed examination of ALM’s and specifically, *SurveyLM*’s potential in a specific, yet critical, area of services research. This approach allows us to highlight the feasibility and relevance of these tools in service research, setting the stage for broader applications and more comprehensive research in the future.

The outcomes of this research are significant for two main reasons. First, they enable cost- and time-efficient comparative analyses of human and synthetic data from notable studies in the field of services marketing, thereby evaluating and calibrating the accuracy of ALM-based agents in service-related research and applications. Second, and perhaps more importantly, this study introduces innovative methods and tools to the field of services research. These advancements have the potential to extend their utility beyond academic research, finding practical applications in market research, user experience testing and other professional domains. The implications of this shift are far-reaching, indicating a potential paradigm change in how complex human behaviors and contextual nuances are approached, analyzed and understood. In this evolving landscape, ALMs (and the large language models or LLMs they are built upon) emerge as important empirical tools, offering a novel means to decode and interpret the fine-grained signals in human interactions and decision-making processes.

This research therefore not only contributes to the existing body of knowledge but also paves the way for future explorations in understanding human behavior through advanced technological lenses. The realm of service research is increasingly investigating the types of service offerings and tasks that are best suited to the incorporation of Artificial Intelligence (AI) to enhance customer experiences (Huang and Rust, 2021). Yet, there is still limited research on the methodological implications of ALMs in service design and innovation, let alone addressing the “replication crisis” through comparative studies. This gap underscores the need for more extensive exploration of how ALMs can be integrated into service research methodologies to ensure the reliability and robustness of findings.

The rest of this paper is structured as follows: Section 2 provides the background literature review, covering a short and focused overview of the history of AI, generative models, and AI agents as research participants. Section 3 details the methodology, including data collection, study selection and the simulation process. Section 4 presents the results, focusing on descriptive analysis and examining the similarities and differences between the original and synthetic studies using SurveyLM. Section 5 discusses the theoretical contributions, managerial implications and limitation, with concluding remarks in Section 6, followed by acknowledgements and references. Additional detail on the methodology, including data extraction procedures, synthetic agent profiles and simulation parameters, are elaborated in the *Online Supplementary Materials* (Bickley et al., 2024), but excluded from this paper for brevity. These materials provide additional insights into the study design, including the structured tables of agent characteristics and other simulation inputs, are available at <https://osf.io/b4udp/>.

2. Background literature

2.1 Short history of Artificial Intelligence (AI)

Over the past few years, LLMs (Zhao et al., 2023) and their augmented counterparts (i.e. ALMs) have gained prominence (Mialon et al., 2023) as powerful tools for research and industry (Grossmann et al., 2023; Korinek, 2023; Wang et al., 2023). These models not only serve as analytical instruments/probes (e.g. to classify text or perform semantic analysis) but also

increasingly comprise the focus of investigation (Aher et al., 2023; Almeida et al., 2023; Arora et al., 2023; Binz and Schulz, 2023; Miotto et al., 2022). This is particularly prevalent in emergent disciplines like machine psychology (Hagendorff, 2023; Simmons and Hare, 2023), but LLMs/ALMs are also being increasingly applied to study problems in the empirically-driven social and behavioral sciences (Holtzman et al., 2023). However, it is important to consider the broader evolution of the field before focusing exclusively on the latest focus or trends in AI development, investment and hype, such as generative AI and LLMs. The evolution of AI has been marked by significant milestones, reflecting a shifting (yo-yo-like) focus between symbolic, logic-based reasoning methods and connectionist, Machine Learning (ML) models and techniques (for more detailed discussion, see, e.g. Bickley and Torgler, 2023). Put simply, AI as a field (i.e. along with related fields like statistics, econometrics, etc.) has developed numerous methods that can be used for a diverse and wide range of real-world problems. But the integration of these tools and methods in an efficient and suitable way is a process that requires exploration and testing.

In business, innovation and entrepreneurship, AI has become a key driver in overcoming barriers to innovation and productivity by enhancing information processing and enabling more effective search routines. To fully realize the potential of AI, it is crucial to identify areas where AI can augment, extend or potentially displace human decision-making. Haefner et al. (2021) highlight, for example, that AI can support four major functions:

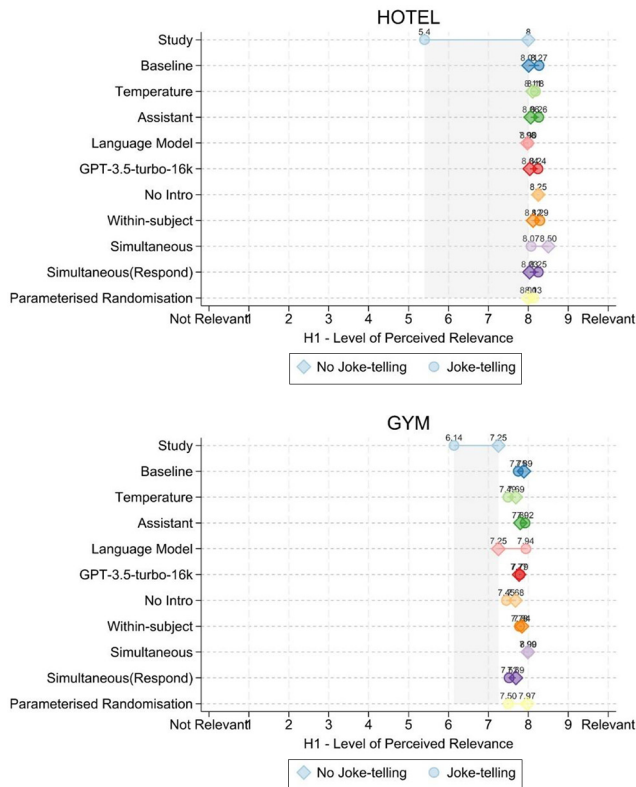
- 1 developing ideas by overcoming information processing constraints;
- 2 generating ideas by overcoming information processing constraints;
- 3 developing ideas by overcoming local search routines; and
- 4 generating ideas by overcoming local search routines (Figure 1 on p. 4).

By augmenting these processes, AI can boost productivity and efficiency, freeing human workers to engage in more creative and strategic activities of higher value/impact. This can lead to more innovative solutions and enhances the ability to address complex challenges in customer, product and market contexts, thereby driving growth and competitive advantage across sectors. Moreover, AI and simulation technologies can significantly enhance strategic planning by providing counterfactuals and scenario planning, addressing asymmetric and incomplete information and enhancing search capabilities across potential pathways (Bickley et al., 2024). Which can enable service firms to better tackle complex, persistent issues and adapt to market changes with greater agility. Some researchers, such as Agrawal et al. (2019) and Goldfarb et al. (2022), suggest that AI could be the next major “general purpose technology” to boost productivity and enable high-impact value creation/translation across a wide range of industries.

2.2 Generative artificial intelligence and (large/ augmented) language models

Generative AI (GenAI) – a subset of AI focused on creating content such as text, images, videos and sound, rather than just processing or analyzing data – has played a more recent role in the current AI (r)evolution. This field leverages a broad range

Figure 1 SO2018 H1 – mean value comparison for joke telling (circles) vs nonjoke telling (squares), including the two scenario variations: HOTEL (top) and GYM (bottom)



Note: See Online Supplementary Materials for detail

Source: Authors' own work

of AI theories and techniques, including Natural Language Processing (NLP), neural networks, supervised and unsupervised ML, Deep Learning (DL) and Reinforcement Learning (RL). GenAI models typically require vast amounts of data for training (e.g. terabytes to petabytes), except for small language models like MobileBERT, Mistral's 7B and Google's Gemma, which are designed to be lightweight and resource-efficient.

Language models are AI systems designed to interpret, generate and translate human language. They have evolved from rule-based systems and knowledge engineering to ML algorithms, with state-of-the-art models now using DL and RL techniques. Examples include OpenAI's GPT models, Google's Gemini models, Meta's Llama models, Cohere's Command models and Anthropic's Claude models. [2] The role and impact of LLMs across diverse fields have expanded significantly, for example, in biomedical and healthcare research and practice (Thirunavukarasu et al., 2023), education, social care, business and agriculture (Raiaan et al., 2024, pp. 26862–9; Hadi et al., 2023; Eloundou et al., 2023), to name a few. LLMs are deep neural networks that predict the next sequence given a current sequence (e.g. of text, image, sound), leveraging vast data and computational power (Dodge et al., 2022; Luccioni et al., 2024). This process, known as inference, involves feeding a sequence of words into the

pretrained model, which then predicts the next sequence of words based on learned patterns. [3] With training and fine-tuning, the base models can perform a wide range of downstream tasks from content analysis and scene recognition to applications like question-answering, summarization and content creation.

LLMs excel in scalability and efficiency during model training and support a wide range of NLP tasks. Their “plug and play” nature and ability to simultaneously encode positional, semantic and syntactic information enhance their portability and practical application. ALMs extend the capabilities of pretrained LLMs by integrating various augmentation techniques such as *prompt engineering* (and chaining), *search and retrieval mechanisms* (function calls and tool use of, e.g. external APIs or application programming interfaces, semantic/vector databases and other local or web-based application plug-ins), *RL* (e.g. from human feedback or for agentic design/development), fine-tuning and integration with other foundational models (multi-agent systems and interaction) and modes of communication (e.g. codex, graph modeling, vision with image and video, audio/sound and voice). These augmentations enhance ALMs' performance in specific knowledge domains and applications (e.g. legal or medical question-answering, chatbots and customer service agents, robotic action planning) by allowing more customized prompts and contextually grounded responses (i.e. greater control over the model's output). They also help address common limitations of pretrained LLMs, such as limited context size, noise filtering, nonfactual responses or hallucinations, biased training data and outdated knowledge.

2.3 Artificial intelligence agents as research participants

LLMs and ALMs may sometimes exhibit remarkable *emergent properties* (Wei et al., 2022; Schaeffer et al., 2024; Michaud et al., 2024), enabling them to perform tasks they were not explicitly trained for, such as higher-level reasoning skills. Thereby, demonstrating a narrow form of generalized intelligence. Trained on extensive human-generated data, ALMs not only display these emergent properties but can also generate human-like traits and behaviors (Hagendorff, 2023). This is evident in their production of human-like explanations (Zhang et al., 2023), impersonations (Salewski et al., 2023), formation of human-like transmission chains (Acerbi and Stubbersfield, 2023) and responses to persuasion and influence (Griffin et al., 2023). Intriguingly, LLMs/ALMs may sometimes also mimic human cognitive processes, such as concept and knowledge organization (Gammelgaard et al., 2023) and reasoning by analogy (Webb et al., 2023).

Researchers are increasingly using ALM- and LLM-powered agents, or “silicon samples”, to conduct empirical research in the social, behavioral, cognitive and psychological sciences with synthetically generated data (Argyle et al., 2023; Dillion et al., 2023; Horton, 2023; Lee et al., 2023; Santurkar et al., 2023). Synthetic data generation involves constructing artificial data that mimics real-world data while preserving its statistical properties, thus protecting sensitive information. This technique is valuable for data privacy, data availability and model training/testing, especially in fields where traditional data collection is slow or difficult. Two important AI-based

approaches for generating synthetic data are federated learning and LLMs. Federated learning (Little *et al.*, 2023) creates synthetic data by learning from decentralized data sets without the need to share the actual data, useful in scenarios where data centralization is not feasible due to privacy or logistical issues (e.g. legal, health, finance). LLMs and ALMs have emerged as a viable alternative even more recently (Eigenschink *et al.*, 2023; Figueira and Vaz, 2022; Jordon *et al.*, 2022; for a good overview, see, Simmons and Hare, 2023). For a detailed overview of the applications of subpopulation representative models (SRMs) using LLMs, see Simmons and Hare (2023). Specifically, the authors review SRMs as models that approximate subpopulation characteristics or aggregate subpopulation sentiment and discuss behavior elicitation techniques for LLMs, existing SRM implementations and frameworks for the analysis, development and practical implementation of LLMs as SRMs.

LLMs generate synthetic data through two main approaches: fine-tuning and prompt engineering. Fine-tuning involves adjusting the model's weights using diverse human-generated data (e.g. real-world responses from the World Value Survey) to enhance performance in specific areas. This method learns patterns within the data, increasing the likelihood of reproducing them, as discussed by Chu *et al.* (2023), Feldman *et al.* (2022) and Kim and Lee (2023). On the other hand, prompt engineering guides a pretrained LLM to produce desired outputs without altering its weights, leveraging the model's general reasoning and knowledge; potentially enabling a more authentic reflection of the human values and biases underlying the model's training and parameters. Both methods have their pros and cons: fine-tuning (in-text prompting) can risk overfitting (underfitting) to persona demographics, whereas prompt engineering may require more careful management and consideration of system-level biases, safety mechanisms and the related choices around model architecture and training.

Using synthetic agent populations in research studies provides several advantages (e.g. see recent survey by Wang *et al.*, 2024). Just as ML has revolutionized materials engineering and drug discovery by enabling the “explor[ation], in silico, large chemical spaces that can be prohibitively expensive for traditional experimental approaches” [4] – state-of-the-art GenAI models could potentially transform service research. By analogy, these AI-driven agents can allow service researchers to explore large product-market spaces that are too costly to investigate using traditional methods. This shift is akin to how ML-based simulations have streamlined drug discovery by identifying promising candidates and minimizing adverse side effects before clinical trials, thereby reducing reliance on trial and error (e.g. with animal trials) and greatly improving efficiency by front-loading the drug design/development process. ALMs make agent-based simulations more accessible for qualitative, qualia-based empirical research, which dominates not only the service and marketing fields but also finds relevance in the empirically driven social and behavioral sciences more broadly. For example, these agents can help researchers/practitioners simulate whether Campaign A or B is received better, whether the intended message is understood, or if there are unforeseen alternative interpretations. They can also help identify what specific aspects agents, e.g. representing

minority populations or target groups/personas A and B, like or dislike.

3. Method

3.1 Study selection

Although the application of LLMs holds the potential to yield significant insights across various aspects of service experiences, this study began with an initial focus on “Joking with customers in the service encounter has a negative impact on customer satisfaction: Replication and extension” by Söderlund and Oikarinen (2018), which itself is a replication study [5]. This paper, herein referred as SO2018, concentrates on the behaviors of frontline employees and their corresponding impacts on customer satisfaction during service encounters. Given the broader focus of this investigation being on the question of replicability using ALMs, SO2018 provided a focused and relatively simple experimental setup with fewer variations, specifically a 2×2 factorial design. This approach allowed us to initially concentrate more intensively on exploring the variations in simulation inputs to the ALMs, such as prompt and model-based adjustments (see Table S5 in the *Online Supplementary Materials*), before attempting more complex experimental designs.

Building on this initial approach with SO2018, we then focused our search on more recent papers published in the *Journal of Services Marketing* during the period from 2022 to 2023. We initiated the search using Scopus with the following criteria:

- **Source Title:** *Journal of Services Marketing*
- **Keywords:** “Scenario” or “Vignette”
- **Publication years:** Between 1993 and 2023

This search returned 92 documents as of 24 June 2024. To ensure relevance and variety, we screened the most recent papers, ultimately selecting four papers from the 2022 and 2023 period. The selection was driven by the structural elements of the methods used in these studies, such as the design and availability of scenario-based experiments, rather than solely on their significance within the field of service research. Note, not many studies shared the actual instructions/scripts used by study facilitators when running the experiments (e.g. to introduce the research, explain the experiments, etc.). Moreover, we had requested the actual code and data used for the analysis in the paper from each of the study's authors (given rather late in the peer review process) but had only heard back from the authors of one selected study (of five) at the time of writing.

Our primary focus was on studies using scenario-based tasks, as these are particularly suited for testing the adaptability and depth of ALMs in mimicking real-world decisions. The selected studies span various sectors/settings and cover different stages and aspects of the service failure and recovery lifecycle. These studies include SO2018 (Söderlund and Oikarinen, 2018) focusing on joke-telling in service encounters, SJS2022 (Srivastava *et al.*, 2022) on misbehavior contagion in access-based services, DGLD2022 (Dootson *et al.*, 2022) on service robot guardians in financial services, YSO2023 (Yan *et al.*, 2023) on unethical pro-organizational behavior in frontline service employees and TLT2023 (Tariq *et al.*, 2023) on intercultural service encounters in the food service sector.

Table S2 in the *Online Supplementary Materials* compares the key differences and aspects of each selected study in terms of service research context, niche/focus, aspect/element of service failure and recovery explored and key findings and considerations for our comparative study.

3.2 Data collection

3.2.1 Synthetic data generation with SurveyLM

SurveyLM leverages in-context prompting of LLMs (specifically, we examined OpenAI's gpt-3.5-turbo in this paper) using synthetic agent profiles matching the demographics of original human study participants. This involves embedding questions, answer instructions and relevant agent profile characteristics within prompts to guide model responses. Although our choice of *SurveyLM* stems partly from convenience, its large-scale response handling features made it superior to alternatives that we explored, such as OpinioAI and Synthetic Users. [6] Compared to Park *et al.* (2023) generative agents or social simulacra, SurveyLM provides a more scalable and cost-effective solution for customizable agent simulations, albeit through a less stylized and immersive display interface.

SurveyLM excels in generating synthetic data for panel-like research and allows method-agnostic approaches to empirical studies of synthetic data generated by ALM-powered agents. It supports fully customizable agent profiles and can run concurrent simulations of hundreds to thousands of agents in a cost- and time-effective manner. Users can generate synthetic agents or upload real-world profiles via CSV/XLSX data sets, [7] enabling the simulation of complex social systems from individual behaviors to collective group dynamics. [8] This flexibility allows researchers to use any empirical or experimental methods, beyond social studies, surveys, qualitative research or interviews. The only input structure required by SurveyLM is a “question” and “answer instruction” column, which serves as the stimulus and response instruction or options available.

The efficiency of SurveyLM is evident when compared to traditional human-centric research methodologies. Tasks that would take days or weeks with human participants can be completed in minutes using SurveyLM. For example, in SO2018 Study 1's HOTEL scenario, engaging 100 agents to respond to 16 questions typically takes less than 10 min. Preliminary analyses show ALM-based research can be 10–20

times more cost effective than traditional human-based research (e.g. using online panels). The total cost for the final simulations in the SO2018 analysis was US\$38.03.

SurveyLM's approach allows structured yet adaptable and cost-effective exploration of ALMs, eliminating the need for technical expertise in API interaction or complex prompt structures. This facilitates a deeper understanding of emergent behaviors and potential applications in services marketing research and other fields by increasing the accessibility to ALM-based simulation studies. The blend of time and cost efficiency achievable with SurveyLM establishes a promising model for integrating ALMs into various research and industry domains. SurveyLM's focus on large-scale simulations with diverse synthetic agent profiles suggests a balanced approach between comprehensive inquiry and operational efficiency, essential in fields requiring large data sets and timely analysis. For more detailed information, refer to the SurveyLM whitepaper (Bickley *et al.*, 2023) and Section 3 of the *Online Supplementary Materials*.

3.2.2 Procedure for generating synthetic data

Generating synthetic data with SurveyLM involves several steps to ensure the accuracy and relevance of the data produced, closely aligning with the original studies' designs and contexts.

Creating synthetic agents: SurveyLM generates synthetic agents by embedding them with attributes reflecting the original sample population. This includes demographic characteristics, behavioral traits and other relevant variables. Initially, we reviewed the selected studies to extract key information on the sample population characteristics such as age, gender, cultural background, socioeconomic status and sample size. The agent profiles are organized in a structured table, with column headers representing the agent characteristics (e.g. age, gender, income level, etc.) and individual agents as the rows. For each study, we populated this table based on the reported demographics (e.g. mean and standard deviations of age, percentage of male and female participants). All other profile parameters (e.g. personality, religion, country of birth/origin, etc.) other than those explicitly described in the original studies were not included in the agent profiles [9]. See Table 1 for an example of the agent profile context for SO2018 Studies 1 and 2.

Extracting survey questions and variables: We also extract the survey questions/items, texts for the scenario and

Table 1 Comparative overview of agent profile characteristics in SO2018 Study 1 and Study 2

Category	Study 1 details	Study 2 details
Original study	$n = 80$, mean age = 24.36, 28 men and 52 women (35% male, 65% female), recruited from business administration courses	$n = 88$, mean age = 20.86, 40 men and 45 women, three participants did not report their gender (45.45% male, 51.14% female, 3.41% did not report gender/prefer not to answer), recruited from business administration courses
Age	{18, 19, ..., 28, 29, 30} years old	{18, 19, ..., 23, 24} years old
Gender	{Female; Male; Other} with probabilities {0.649, 0.349, 0.002}	{Female; Male; Undisclosed} with probabilities {0.5114, 0.4545, 0.0341}
Education	A student enrolled in at least one business administration unit	A student enrolled in at least one business administration unit

Source: Authors' own work

treatment conditions and other key variables from the selected studies. This involves a detailed review of the studies to gather information on the experimental design, including independent, dependent, mediating and moderating variables. The extracted survey inputs need to be formatted according to SurveyLM's specifications, i.e. a structured table with columns for "question" "question id" and "answer instruction". This ensure that the synthetic data aligns closely with the original study designs. For the other four studies, this table included all survey questions/items, scenarios, manipulation checks and any other additional variables (e.g. demographics), presented in blocks of questions (e.g. all demographic questions together and survey questions grouped according to the multi-item measures they proxy) in the order that could be inferred from the original studies. For SO2018, we excluded any demographic questions and tested the model responses when presented with the questions as a single row/block of questions. This did not qualitatively change the results.

Running simulations: After setting up the synthetic agents and survey questions, we run SurveyLM simulations to generate data. [10] The simulations mimic the conditions and contexts of the original studies, producing synthetic data for comparison with the original human data. Key model parameters, such as model temperature range, specific LLM model, agent roles and their probability distribution. The full set of model parameters and their definitions and some example use cases are provided in Table S3 of the *Online Supplementary Materials*. We then estimate the simulation cost and run the simulations.

The number of agents is based on the sample population from the original study. For SO2018 Studies 1 and 2, we aimed to roughly double the original sample sizes (Study 1: $n = 80$; Study 2: $n = 88$) and ensure a balanced distribution of agent response counts across each of the 40 variations (see Table S6 in the *Online Supplementary Materials*) explored with the SO2018 study. [11] This led to analyzing 5,122 agents from simulations run between November 13 and 20, 2023, which is about 30 times the original sample size. [12] For the other four studies, we aimed for a minimum of 500 agents per experimental treatment/variation introduced in each study. This second set of simulations were run between July 1 and 12, 2024.

3.3 Analysis and evaluation

To maintain a focused and manageable scope for this initial empirical investigation of ALMs within the field of services marketing, we chose to concentrate our analysis on the primary manipulation checks and hypothesis tests of the original studies. This decision was guided by the need to succinctly address the most critical aspects of the original research, allowing for a more efficient yet thorough (conceptual and operational) comparison to the original study's key findings. We precisely followed the original study's method and procedural instructions, using the same statistical methods, tests and procedures used to generate results for primary manipulation checks (e.g. M1, M2) and hypothesis tests (e.g. H1, H2) as reported in the original studies. By mirroring these methods as closely as possible, our evaluation provides a more direct comparison between the findings generated from ALM responses in this study and the human-generated findings from the original study.

As mentioned in Section 3.1 (Study Selection), we focus on the SO2018 study first. We start by comparing the results of our synthetic study with those established in the original study, showing the means differences of the study's main manipulation checks (M1, M2) and hypotheses (H1, H2). This is a crucial step to ascertain the degree of alignment between our SurveyLM findings and the original study. Below this, we discuss briefly two graphs that show the comparative results for all nine prompt- and model-based variations we tested on H1 and H2, respectively. The synthetically generated results are presented alongside the original study's results to allow direct comparisons between the results from the synthetic agents to those of the human participants from the original SO2018 study.

We then conducted a similar analysis for the four studies from the *Journal of Services Marketing* as a robustness/sensitivity check and extension of the results from the first section. By comparing the results from these additional studies, we aimed to further validate the utility and reliability of ALM-generated synthetic data in service research. We precisely followed each original study's method and procedural instructions. In Section 4.2 (Comparative Analysis of the Four JSM Studies), we first present a figure which compares the internal reliabilities of key measures between original human participants and synthetic participants across the four selected studies. We then provide a table that shows the alignment of hypotheses testing results and highlights any key differences and similarities that we observed when looking at the detailed numerical results from each study. Detailed results of these additional studies, including a more detailed account and report of the numerical measures for each of the studies and the model- and prompt-based variations we tested, are provided in the *Online Supplementary Materials* to maintain brevity and focus in the main text.

4. Results

4.1 Comparative analysis of the SO2018 study

In this section, we undertake a thorough comparison between the formal results of SO2018 and the responses obtained using SurveyLM. In SO2018, the treatment manipulation checks were designed to assess whether the experimental treatment, namely, joke-telling, was perceived similarly by the human participants as intended. Similarly, the hypotheses:

- H1. Employee joke-telling in face-to-face interactions with the customer reduces the customer's perceptions of the relevance of what the employee says (p. 7).
- H2. Employee joke-telling in face-to-face interactions with the customer reduces customer satisfaction (p. 8).

were formulated to test specific outcomes of the joke-telling intervention. In both studies (HOTEL and GYM scenarios), the original SO2018 study suggested the manipulation behaved as intended, in which the jokes were identified or perceived as fun in the treatment condition. Similarly, H1 and H2 were supported. Table 2 juxtaposes the findings from SO2018 with those derived from the SurveyLM.

First and foremost, we note that M1 and M2 are supported in the Hotel study. SurveyLM agents exposed to the joke-telling condition were more likely to perceive that the employee

Table 2 Comparative analysis of SurveyLM and SO2018 study outcomes in the HOTEL and GYM scenarios

	Hotel (SO2018)						Gym (SO2018)				
	Mean (Joke)	Mean (No joke)	Diff	<i>t</i>	<i>p</i>		Mean (Joke)	Mean (No joke)	Diff	<i>t</i>	<i>p</i>
M1	6.88	2.59	4.29	8.3	<0.01	M1	7.62	3.02	4.6	10.02	<0.01
M2	7.07	5.9	1.17	2.89	<0.01	M2	7.54	6.25	1.29	3.18	<0.01
H1	5.4	8	−2.6	7.05	<0.01	H1	6.14	7.25	−1.11	3.4	<0.01
H2	6.79	7.5	−0.71	2.11	<0.05	H2	6.25	7.03	−0.78	2.27	<0.05
	Hotel (SurveyLM)						Gym (SurveyLM)				
	Mean (Joke)	Mean (No joke)	Diff	<i>t</i>	<i>p</i>		Mean (Joke)	Mean (No joke)	Diff	<i>t</i>	<i>p</i>
M1	5.93	5.06	0.87	9.7	<0.01	M1	8.23	7.58	0.65	7.45	<0.01
M2	7.87	7.05	0.82	14.7	<0.01	M2	8.49	8.79	−0.3	6.44	<0.01
H1	8.27	8.01	0.26	18.3	<0.01	H1	7.65	7.89	−0.24	8.9	<0.01
H2	7.12	7.95	−0.83	25.08	<0.01	H2	7.23	7.27	−0.05	2.19	<0.05

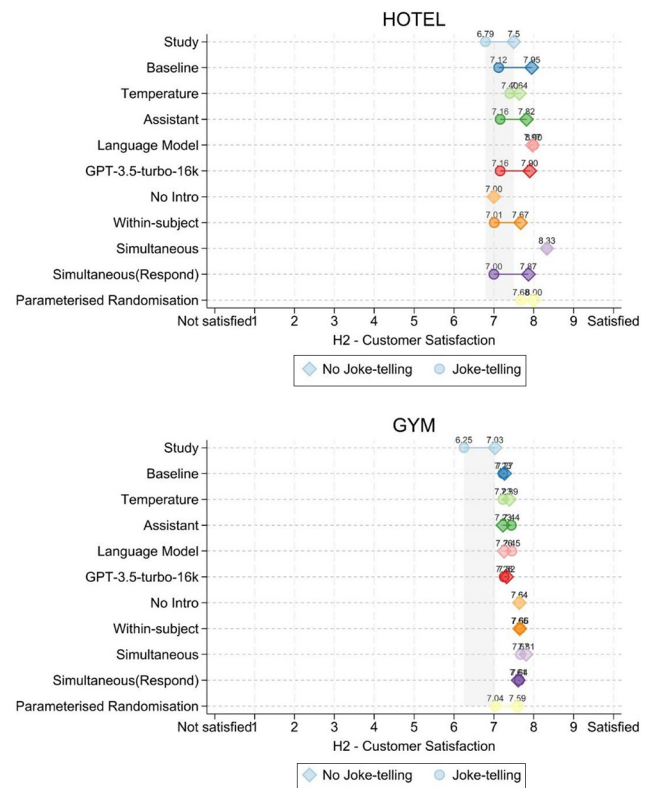
Notes: This table provides a detailed comparison between the results of the SurveyLM baseline variation and the original SO2018 study, focusing on the perceived impact of joke-telling in both Hotel and GYM study scenarios

Source: Authors' own work

told a joke and found the conversation to be more fun. However, in the GYM study, while M1 was also supported, indicating the recognition of joke-telling, the agents did not perceive the instructor as more fun in the joke-telling condition compared to the no joke-telling condition. Perhaps, the words “fun” and “gym [instructor]” rarely occur close together in gpt-3.5-turbo’s training data sets. It is also noteworthy that the mean differences for M1 in SurveyLM are notably smaller, approximately five to seven times less, compared to SO2018.

When testing *H1* and *H2*, the results from SurveyLM showed different levels of support compared to SO2018. In the HOTEL no-joke-telling condition, the average perceived relevance was similar (8 vs 8.01), but SurveyLM agents rated the perceived relevance higher in the joke-telling condition, contrary to the lower value of 5.4 found in SO2018. Although *H1* was supported in the GYM study using SurveyLM agents, the mean difference was about 4.6 times smaller than the original human-based study. Support for *H2* was found in the SurveyLM baseline variation for both the HOTEL and GYM studies. Although the *H2* mean difference was similar in size to that in SO2018 for the HOTEL scenario, in the GYM study the mean difference was substantially smaller, being 15.6 times less than that found in SO2018. One possible reason for these observed differences includes the limitations of written text in capturing complex human emotions, social interactions and the contextual nuances of hypothetical scenarios. More specifically, the inverse relationship for *H1* in the HOTEL scenario could indicate a general preference or expectation for the contextual use of humor in hotel settings (i.e. which the LLMs have learnt during training). More generally, this finding highlights potential differences in how humor is interpreted by the AI model versus human participants.

In Figures 1 and 2, we visually illustrate the mean values for M1, M2, *H1* and *H2* under the full set of model and prompt-based treatments variations that we tested. The general story of these figures is the synthetic study provides much smaller estimates of the treatment differences than the original SO2018 experiment in both scenarios. The majority (90%) of the variations for the HOTEL scenario resulted in a small positive treatment difference, clearly contradicting the SO2018 findings

Figure 2 SO2018 *H2* – mean value comparison for joke telling (circles) vs nonjoke telling (squares), including the two scenario variations: HOTEL (top) and GYM (bottom)

Note: See Online Supplementary Materials for detail

Source: Authors' own work

for *H1*. However, six out of 10 variations for the GYM scenario provided evidence in support of *H1*. For *H2*, seven and five out of 10 of the variations for the HOTEL and GYM scenarios (respectively) provided results that converged with the direction of the means difference between the joke/no joke treatments in the original study. But again, treatment

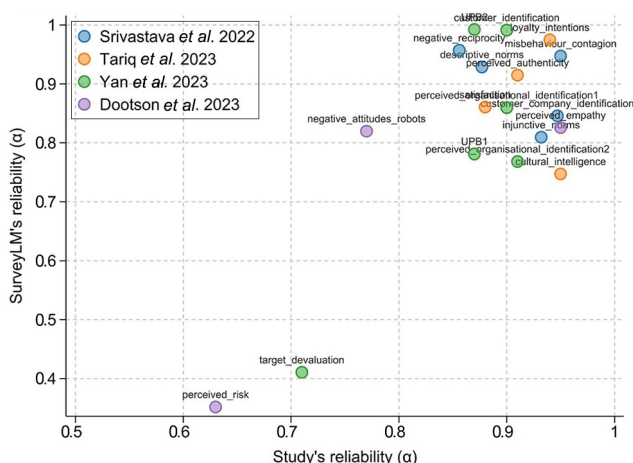
differences were all smaller than those reported in the original study.

For a more detailed analyses of these variations, please refer to Sections 4–6 in the *Online Supplementary Materials*. Specifically, Table S5 includes definitions of the nine different model- and prompt-based treatment variations we tested, including the “Baseline” settings, which Table 2 adopts. For instance, see Section 5 in the *Online Supplementary Materials* (Bickley et al., 2024), where we examine the influence of model temperature and its relationship to agent response variation, finding on average that higher temperature settings in ALMs led to greater variability in responses. This finding suggests the stochasticity introduced by increasing model temperature can significantly impact the diversity of ALM responses, meaning they can be fine-tuned to be (marginally) better at mimicking the spontaneity and randomness of human responses. We also provide verification checks that our synthetically generated population of the agents and their demographics (i.e. age, gender, and education) accurately fit the specific parameters we intended and mirrored the demographic profiles reported in the original SO2018 study. Specifically, the summary statistics for age and gender reported.

4.2 Comparative analysis of the four Journal of Service Marketing studies

Moving onto the comparative analysis of the selected studies from JSM, Figure 3 summarizes the correlations between the measurement construct internal reliabilities for key measures and manipulation check variables as generated by the original human participants compared to the synthetic participants across the four studies. Overall, the SurveyLM agents’ responses to the study items (questions) correspond quite closely to how human participants in each study does. The level of reliability of the measures corresponds well to that produced by SurveyLM agents, with a correlation of 0.758 between the two sets of reliabilities. For lower values of the original study’s

Figure 3 Comparative analysis of the internal reliabilities of key measures and manipulation check variables between original human participants and synthetic participants (SurveyLM agents) across four selected studies



Source: Authors’ own work

reliability along the x -axis, such as with “perceived_risk” and “target_devaluation” constructs, it is worth noting that SurveyLM shows an even lower reliability measure, relative to the human sample. This indicates that SurveyLM and synthetic agents might help map reliability/internally consistent measurement constructs (e.g. multi-item scales to their overall construct they are measuring) and help identify which measures are scoring low. For example, if it scores low for both a human and a synthetic agent, then perhaps the measurement construct should be reconsidered. We include Table S12 with the actual values of reliability in the *Online Supplementary Materials*, along with the number of items for each measure in the original study.

Table 3 shows whether the hypotheses testing results of the original studies were reproduced with the response obtained from SurveyLM agents. Key findings from the comparative analysis indicate a mix of alignment and deviation between the synthetic- and human-generated data. Of the 30 hypotheses tested, 10 were supported, four were partially supported and 16 were not supported. Detailed numerical results for each study are again deferred to the *Online Supplementary Materials*.

5. Discussion

Overall, our comparative analysis reveals both strengths and limitations of using synthetic agents to replicate human-based empirical studies in services research.

The alignment for M1 and H2 in Söderlund and Oikarinen (2018) shows that synthetic agents can effectively detect and respond to clear and straightforward prompts. The partial alignments for M2 and H1 suggest that the AI might struggle with more context-dependent evaluations like the perception of fun or relevance, which could be influenced by subtleties in the social setting or individual humor preferences. The variance observed in Study 2 (GYM) could be due to differences in the setting, as humor might be perceived differently in a gym versus a hotel context. In addition, the AI’s training data might not fully capture the intricacies of social interactions in these specific environments, leading to discrepancies in more nuanced assessments.

For Dootson et al. (2022), none of the hypotheses were supported, with all showing effects with opposite signs. This discrepancy could be due to the primary reliance on visual images for treatment manipulations, which SurveyLM cannot replicate as it only supports text-based inputs at the time of writing this paper. Another possible reason could be an emergent assumption in the gpt-3.5-turbo model, suggesting that humans tend to cheat other humans more than robots, as potentially indicated by the training data. In addition, synthetic agents might struggle with iterative and second/third-order thinking, affecting their responses to questions involving multiple perspectives, as seen in the DGLD2023 scenarios (i.e. asking what the hypothetical “Sam” would do).

For Yan et al. (2023), H1 showed an effect with the opposite sign, indicating challenges in capturing the dynamics of unethical pro-organizational behavior. One reason for this discrepancy could be the lack of an explicit column in the agent profile context for “Organizational Identification”, as reported in the original study (mean and standard deviation data for Studies 1a, 1b and 2). Although agents were asked to self-identify their organizational identification measures, high

Table 3 Summary of hypothesis testing results with definition of the hypotheses and including the original study findings

Study*	Hypothesis	Definition	Original findings	Alignment with study	Reason
Dootson et al. (2022) (DGLD2023)	H1	"As the perceived humanness of the service agent increases, intentions to commit deviant consumer behavior decrease". (p. 279)	Supported	No	Opposite sign
	H2	"Empathy mediates the negative relationship between perceived humanness of the service agent and deviant consumer behavior intentions". (p. 279)	Supported	No	Opposite sign
	H3	"Perceived risk mediates the negative relationship between perceived humanness of the service agent and deviant consumer behavior intentions". (p. 279)	Supported	No	Opposite sign
	H4	"Negative attitudes towards robots will moderate the relationship between perceived humanness of the service agent and deviant consumer behavior intentions, such that negative attitudes towards robots will strengthen the negative relationship". (p. 279)	Supported	No	Opposite sign
Yan et al. (2023) (YSO2023)	H1	"Employees' organizational identification positively relates to unethical pro-organisational behavior (UPB) only when there is a low level of customer identification". (p. 929)	Supported in Studies 1a, 1b and 2	No	Effect with opposite sign to all three studies
	H2a	"The interactive effect of organizational identification and customer identification on UPB will be mediated by target devaluation. That is, organizational identification will positively affect target devaluation only when there is lower customer identification, . . ."	Supported in Study 2, not tested in Study 1	Partial	Same sign, but not significant
Srivastava et al. (2022) (SJS2022)	H2b	and target devaluation, in turn, positively affects UPB". (p. 930)	See previous	Partial	Same sign, but not significant
	H1	"Previous misbehavior of customers in access-based services leads to misbehavior contagion among other customers". (p. 851)	Supported	No	Insignificant effect
	H2	"Previous misbehavior by other customers leads to an increased perception of descriptive norms by the focal customer" (p. 851)	Supported	Yes	
	H3	"Perception of descriptive norms by the focal customer mediates the effect of previous misbehavior by other customers on misbehavior contagion by the focal customer" (p. 851)	Supported	No	Opposite sign
	H4	"A focal customer's perception of injunctive norms moderates the mediating effect of descriptive norms: The mediation effect is weaker when the focal customer's approval of the injunctive norm of using the accessed good responsibly is stronger". (p. 852)	Supported	Yes	

(continued)

Table 3

Study*	Hypothesis	Definition	Original findings	Alignment with study	Reason
Tariq et al. (2023) (TLT2023)	H5	"Interpersonal anonymity moderates the relationship between the previous misbehavior and misbehavior contagion: The effect of the previous misbehavior on misbehavior contagion is weaker in a non-anonymous condition than in an anonymous condition" (p. 852)	Supported	No	Opposite sign
	H6	"CCI moderates the relationship between misbehavior and misbehavior contagion: The effect of the previous misbehavior on misbehavior contagion is weaker for customers with high CCI than customers with low CCI" (p. 852)	Supported	No	Opposite sign
	H1a	"(a) Satisfaction with the service experience and (b) loyalty intentions are higher for moderate (than low or high) levels of adaptation." (p. 959)	Supported in Studies 1 and 2	Yes	
	H1b	See above	See above	Yes	
	H2a	"Perceived authenticity mediates the effect of the level of adaptation on (a) satisfaction and (b) loyalty intentions." (p. 959)	Supported in Studies 1 and 2	Yes	
	H2b	See above	See above	Yes	
	H3a	"Generational cohort moderates the effect of adaptation level on (a) satisfaction and (b) loyalty intentions such that compared to younger respondents, older respondents show higher (lower), satisfaction and loyalty intentions at high (low) adaptation" (p. 960)	Not supported in Study 1 (significant in opposite direction), not tested in Study 2	Yes	
	H3b	See above	See above	Yes	
	H4a	"Customers' CQ will moderate the relationship between the level of adaptation of the service offering and the outcome variables of (a) satisfaction and (b) loyalty intentions such that high levels of CQ will mitigate the effect of low and high levels of adaptation on satisfaction and loyalty intentions." (p. 965)	Not tested in Study 1, supported in Study 2	NA	Unable to reproduce
	H4b	See above	See above	NA	Unable to reproduce

Source: Authors' own work

homogeneity in responses (due to temperature = 0) typically led to lower organizational identification levels compared to the real-world study. In addition, there was a general bias of the model towards “good” and “moral” actions. If we had directly inputted different levels of organizational identification, we might have seen higher sampling of such attributes. Conversely, original human participants might have overstated their organizational identifications (e.g. due to social desirability and courtesy biases and other demand characteristics). *H2a* and *H2b* had partial alignment, suggesting some capability of ALMs in understanding complex social dynamics but highlighting gaps in accurately capturing the context of these nuanced interactions through natural language prompts.

Srivastava *et al.* (2022) presented a varied picture. Although *H2* and *H4* aligned with the original study, other hypotheses showed insignificant or opposite effects. This points to potential gaps in the AI’s training data regarding social norms and behaviors in access-based services. For example, with *H1*, synthetic agents might struggle with understanding nuanced social contagion dynamics without explicit visual cues or context-rich scenarios. The discrepancies could be due to, for example, the limited demographic attributes (only gender and bike usage) and the absence of visual stimuli, which might be essential for replicating the nuanced social and behavioral contexts of the original study.

Tariq *et al.* (2023) showed strong alignment for most hypotheses, suggesting that AI models can effectively capture intercultural service encounter dynamics (at least in the context of dining experiences relating to a local Indian restaurant). However, the inability to reproduce *H4a* and *H4b* indicates that aspects like cultural intelligence and authenticity in service interactions require further exploration.

Although some studies show strong alignment, significant discrepancies in others emphasize the need for further research to enhance the accuracy of AI-generated data in mimicking human responses to complex social contexts. Our findings therefore contribute to the broader literature of AI by demonstrating the context-dependent nature of ALM responses, moving beyond simple benchmarking like task completion to benchmarking the complex social processing and dynamics of ALM-based agents.

5.1 Theoretical and methodological contributions

This paper makes several theoretical and methodological contributions to the services marketing field by presenting ALMs as significant enablers and accelerators of research and service enhancement. By providing a novel method for analyzing service failure and recovery scenarios, this study demonstrates the potential of ALMs to significantly enhance and accelerate research and service improvement. Although existing research has demonstrated how tools such as ChatGPT can impact the service industry across five areas (Sigala *et al.*, 2024), we make a contribution by presenting another potential avenue for generative AI – as a notable methodological tool for service scholars and practitioners alike. We contribute to the literature on ALMs within service research by validating their use in simulating human-like responses in complex service interactions, thus understanding and establishing their reliability, utility and associated issues as generative research tools in this domain. In doing so, this

research highlights the feasibility of using ALMs to explore and understand customer experiences and responses in various service settings, paving the way for broader applications and more comprehensive research in the future. Although we initially present the value of such tools in exploring service failure and recovery efforts, we make our methodology transparent to also allow other subsections within service industries to advance the theoretical understanding of how contextual factors impact consumer experiences.

Although AI tools are appealing due to their promises of improving productivity and objectivity by overcoming human shortcomings, there are also well acknowledged risks and considerations related to implementing AI tools across the research pipeline (Messeri and Crockett, 2024). Namely, using AI tools can exploit cognitive limitations in the sense that they can lead to illusions of understanding, where scientists or practitioners believe they understand more about the world than they actually do – making significant inferences or managerial decisions based on limited knowledge of the underpinning processes within the “black box” of AI and generative tools they employ. As a result, science becomes more vulnerable to errors and certain viewpoints can begin to dominate over alternative approaches. The proliferation of AI tools risks introducing a phase of scientific inquiry where more is produced, but less is understood, emphasizing the need for responsible knowledge production in the age of AI guided also by a good theoretical understanding.

This study contributes to the conversation on the AI-powered service landscape (Sigala *et al.*, 2024) with an initial demonstration of how ALMs can be responsibly integrated into the services marketing field. By validating the use of ALMs to simulate human-like responses in service interactions, this research suggests that ALM’s can be an initial proxy data set to mimic consumer responses to a specific service enhancement. We present several caveats and considerations, such as the importance of understanding model biases, the need for context-rich agent profiles and the influence of other model characteristics like training data quality and temperature settings. By transparently demonstrating the process of comparative analysis across five empirical studies, the black box of generative AI is made more tangible for service scholars. We make a contribution by providing a more robust framework to intelligently use future AI applications in service research, highlighting the importance of understanding customer experiences and responses through ALMs to enhance and accelerate research and service improvement if used adequately. These insights pave the way for broader applications and more comprehensive research, ensuring that the integration of AI tools (e.g. LLM-based chatbots and service agents/assistants) fosters innovation and maintains the integrity of scientific inquiry.

5.2 Managerial implications

The results of this study have practical implications for services management and marketing, providing a promising avenue for exploring how alterations within the dynamics of a service encounter, however small or large, may impact members of the service ecosystem (FLEs, consumers and co-consumers). Services involve high levels of variability and heterogeneity, particularly with multiple actors and social scripts co-occurring

throughout a service encounter. ALMs can be a “first wash” test of managerial strategies to respond to different service situations. The knowledge and insights gained from “rapid” explorations of research ideas via ALMs can have several immediate implications, especially in time sensitive situations. Thereby, acting as an innovation catalyst for service managers and human front line employees.

The use of LLMs and ALMs may also provide a fruitful avenue to explore taboo or “risky” research variables, without the associated reputational risk for the service organization in question. For instance, the Stanford Prison Experiment (Zimbardo *et al.*, 1971) is an (albeit extreme) example of an important human behavior study that would likely not pass today’s ethical standards to allow for replication of experimental design. Similarly, the Milgram shock experiment (Milgram, 1963), which examined obedience to authority, would face significant ethical scrutiny today. In a similar vein, service encounters of a high-risk nature may benefit from initial exploration with artificial agents, such as experimental studies or interventions to address customer verbal aggression or intentions to harm frontline employees in service failure situations (Sommovigo *et al.*, 2019).

Our findings from SO2018, for instance, demonstrate how understanding the variability in ALM responses to humor can help managers tailor customer interactions, ensuring that AI agents are used effectively in contexts where humor and personal touch are essential. Similarly, this understanding ensures that AI agents are not used in contexts where humor or personal touch are unconventional and even perceived as rude or inappropriate. For example, in retail or hospitality settings, AI agents could be calibrated to respond with a degree of humor or formality depending on the customer’s profile or the nature of the interaction. More generally, this highlights the need for front-end design and calibration of ALM agents (e.g. with trial users, focus groups) to better align agent behaviors with human expectations and norms, underscoring the importance of careful management and integration of AI-based customer service interactions into operations, to foster trust and meet customer expectations regarding different contextual elements of the service experience.

Nevertheless, there are significant underlying risks and hazards associated with the application of LLMs and ALMs which require attention, understanding and effective management, as highlighted by research like Weidinger *et al.* (2022), Hendrycks *et al.* (2021) and Messeri and Crockett (2024). Service managers need to establish clear guidelines for the ethical uses of AI, including transparency about AI-driven interactions and respect for customer privacy and sensitive/personal information. It is also important to consider and account for potential empirical differences between human- and AI-driven insights in their service improvement strategies, using real-world customer feedback to further refine AI interactions and service offerings in a continuous process of alignment and extension (e.g. like goal posts and wayfinding in between). It is also important for service managers to account for differences between human and AI insights in their service improvement strategies, using real-world customer feedback as waypoints to continuously refine AI interactions and offerings, leveraging AI models and synthetic data generation to explore and map deeper contextual factors and mechanisms in between.

5.3 Limitations and further research

There are several limitations of our comparative analysis, alongside avenues for further research. Although the potential of ALMs for synthetic data generation has been highlighted, the limitations and risks should also be noted, underscoring the need for ongoing research. Issues, like biased data, reduced diversity of thought and homogenous outputs underline the need for careful and critical interpretation of AI-generated outputs (Weidinger *et al.*, 2022; Hendrycks *et al.*, 2021; Messeri and Crockett, 2024; Breithaupt *et al.*, 2024). For example, although ChatGPT provides competent summaries, human retellers introduce greater novelty through spontaneous creativity (Breithaupt *et al.*, 2024). AI models often also have inherent limitations and biases (e.g. a positive or nonnegativity bias) based on their training data and/or safety-based system-level prompting or (instruction) fine-tuning. If the models were trained on data sets that are not representative or contain biases, this could influence the responses of LLM-based agents, potentially deviating from how real humans with a similar demographic profile and characteristics might respond. Comparing different models from the same (e.g. GPT-4 and GPT-4o) or different LLM system providers (e.g. Anthropic’s Claude 3.5 Sonnet) would provide insights into model performance, especially given their multimodal capabilities, opening possibilities for audiovisual inputs in future studies.

Hagendorff *et al.* (2023) report that some intuitive behaviors and reasoning biases present in earlier LLM iterations seem to have been mitigated in newer models, suggesting continuous learning and adaptation. In other words, relying on specific iterations of language and AI models (such as gpt-3.5-turbo or its variants) means that future iterations, which may undergo changes through processes like reinforcement learning from human feedback (RLHF), could yield different results. Therefore, another natural extension of this research would be to reconduct the analysis at a later point in time, using newer iterations of the AI models. Therefore, allowing to examine the robustness and consistency of AI-generated results over time. Longitudinal analyses will also be instrumental in understanding the dynamic nature of AI model responses through user/consumer feedback (e.g. through fine-tuning and RLHF).

Another key limitation is the lack of actual data to extract one-for-one respondent demographics for synthetic agent profile construction to check respondent-level alignment. Because we can only rely on the reported distribution parameters (e.g. mean and standard deviation) of different characteristics/traits from the original sample, but not the covariation between these observable characteristics, during the profile construction of the synthetic agents, we assume the characteristics assignments are independent (e.g. the covariance matrix has all 0 values). For example, if in the real sample older subjects are more likely to be used (age is positively correlated with employment), our synthetic agents will not reflect that unless this information is provided in the original study. Responses may also be impacted by nonreported characteristics absent in the original study. Including these details would be a great avenue for future studies. Moreover, some studies only discussed significance levels and did not report anything about magnitude of the reported relationship. Future work could prioritize studies with more comprehensive documentation, including detailed methodologies, scripts,

code and data and (anonymized/desensitized) participant information/profiles; before then relaxing these criteria and attempting more complex (and novel) experimental designs. This would enable a more accurate reconstruction of original experiments within the SurveyLM platform. Collaborations with the original study authors could also be sought to ensure fidelity in replication and to gain insights into the more nuanced aspects of replicating the original research design.

Finally, although AI agents can emulate (at least to some degree) different human demographics and behaviors, they may not fully capture the depth and subtleties of the human experience and decision-making processes. For instance, Wang *et al.* (2024) have pointed out the inherent limitations and potential harms of misrepresenting social identities like gender and race. In this paper, synthetic agents are programed based on averages and standard deviations of demographic variables (e.g. age, gender) as typically in experimental work, there is limited time/space for demographic questions during data collection. Although this paper aimed to replicate the original human experiments, contextual differences between AI-generated responses and human behavior, including unobserved demographic variables, culturally relevant information and situational cues, therefore remain. This limitation is highlighted by studies such as Atari *et al.* (2023), which suggest LLM responses often reflect more the perspectives of WEIRD societies that needs to be considered and overcome (e.g. by oversampling minority demographic variables when developing the synthetic agent profiles). Schmidt's (2009) model for replicability proposes four major tiers of replicability: primary information focus (immaterial and material aspects), contextual background (encompassing six subcategories), participant selection and allocation procedures and procedures for constituting the dependent variable (e.g. how variables are measured/data is collected, the physical apparatus involved, analytical methods) – where contextual differences may have influenced the findings of our comparative study. It would be an interesting avenue for future research to explore replicability of human-based research along all four axes in addition to deeper/ richer contextual representations of the human agents being synthetically generated.

6. Conclusions

This study serves an initial and pivotal step in applying ALMs as empirical tools within services marketing and broadly in the field of social sciences. It offers critical insights into the capabilities and constraints of using ALM-based synthetic data generation by comparing human and AI-generated responses to scenario-based experimental tasks/treatments, thus paving the way for future explorations in this emerging domain. Namely, our findings underscore the potential and current limitations of ALMs in accurately mimicking complex human behavior and responses in scenarios/situations of relevance to services researchers.

The findings contribute theoretically by demonstrating ALMs' capacity to interpret and respond to complex social cues like humor (SO2018), or ethical decision-making (YSO2023), showcasing their contextual sensitivity. Practically, the results can guide the application of ALMs in service marketing, customer service and AI-driven interactions, emphasizing the need for role-specific programing, calibration and verification of AI systems.

As ALM technology evolves, further research in this direction promises to enhance our understanding and application of these powerful empirical tools across various domains. We suggest future researchers explore other specific micro-level relationships within the integrative framework for service recovery (Van Vaerenbergh and Orsingher, 2016). In this paper, we focused our efforts on just a few of these relationships. However, the versatility and effectiveness of these processes, as demonstrated in this study, underscore their potential utility in a wide array of academic inquiries. Consequently, they can serve as a valuable template for future research seeking to replicate, validate and expand upon findings in diverse areas of study.

Notes

- 1 ALMs extend the capabilities of pretrained large language models (LLMs) – such as OpenAI's GPT-4o or Anthropic's Claude 3.5 Sonnet – via techniques such as prompt engineering, search and retrieval mechanisms (via APIs, plug-ins and other tool use), reinforcement learning, fine-tuning and integration with other foundation models and communicative modalities like vision, audio and codex. For a detailed review on LLMs, see Zhao *et al.* (2023).
- 2 For an excellent online tutorial (YouTube video) providing an overview of LLMs and ALMs see, for example, the following by computer scientist, Andrej Karthy: https://youtu.be/zjkBMFhNj_g?si=F_9Z94e_N2aPJUY1
- 3 The training of LLMs begins with pretraining, where a large corpus of text (e.g. 10TB) is compressed (in an abstract sense) into the model using extensive computational resources (e.g. 6,000 GPUs over 12 days) to create a base model. This base model is then fine-tuned with high-quality, manually curated data sets, refining the model to respond more appropriately to specific prompts/instructions and thereby creating an assistant model.
- 4 This is a direct quote from Regina Barzilay of MIT's Computer Science and AI Laboratory (CSAIL) for a 2020 MIT press release article, see ref: <https://news.mit.edu/2020/artificial-intelligence-identifies-new-antibiotic-0220>
- 5 The search and selection process for SO2018 was initiated with keyword searches for “scenario-based experiment”, “scenario-based survey”, “scenario-based decision-making task” and “vignettes”, on Google Scholar, Scopus and our institutional library aggregator.
- 6 See websites at www.opinio.ai/ and www.syntheticusers.com/. Another noteworthy alternative, which we are yet to fully explore, is the “homo_silicus” python package developed by Kar (2023).
- 7 This so-called “direct upload” feature, for example, could have been used to directly replicate the demographic variables using the data gathered during the real-world studies, for example, but unfortunately, we did not have access to the data at the time of writing.
- 8 One approach is to reconceptualize the agent from “individual” to “group/team” and create a new “group” role. This allows for formulating prompts that capture collective responses (e.g. votes, donations, opinions) and

examine the group processes, dialogues and interactions leading to these outcomes, providing a rich probe into multi-agent system behaviors and dynamics.

- 9 This is one potential reason that the “collective” synthetic agents’ responses may not well aligned with the ones from the human studies, as the “unobservable” (better yet nonreported) characteristics were not clearly specified in the original studies.
- 10 We used the following simulation settings/parameters for each of the JSM studies: model = gpt-3.5-turbo, batch_survey = True, reset_parameters = False, test_run = False, temperature range 0 (low) to 0 (high) and max_retries = 5. For the SO2018 study, the settings/parameters are reported in the *Online Supplementary Materials* as we tested a number of prompt- and model-based variants in this initial study.
- 11 40 variations in total = 4 vignette-based * (3 model-based + 6 prompt-based + 1 baseline proxy).
- 12 An alternative approach to our study could have involved exploring within-subject variations as the baseline setting (i.e. instead of the between-subject default in the current study) across all model- and prompt-based variations outlined in Table 3, without rerandomizing the agents’ profile context between each simulation round (i.e. effectively keeping the same pool of agent profiles). This method, while reducing the total sample size back to $n = 100$, could potentially offer a more focused and precise analysis on the effects of prompt context and framing (from the perspective of individual agents). We, however, opted for the between-subject design as our baseline setting, aiming to showcase the significant potential of LLMs/ALMs and generative AI in generating diverse and varied large-scale synthetic data.

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Supplementary material

The supplementary material for this article can be found online.

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